

# Zero-Touch Manufacturing: A Fully Automated Semiconductor Wafer Handling Process

## **Group 1**

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# Abstract

In this project, our group presents the design, simulation, and validation of an integrated robotic semiconductor handling cell developed in CoppeliaSim. The system was modeled on a localized wafer-handling workflow in which Front Opening Unified Pods (FOUP) are transported by a conveyor system, transferred by a Kinova Jaco robot, and unloaded by a smaller Niryo One precision robot for wafer alignment and processing. The objective of the simulation is to demonstrate how multiple robotic and mechatronic subsystems can be coordinated to perform a realistic industrial automation task involving material transport, robotic manipulation, machine vision, and machine learning-based orientation correction.

The simulated system is organized around three primary stages. First, FOUPs are delivered to the work cell using a floor conveyor that functions as both a transport mechanism and a queuing buffer. Second, the Jaco robot, referred to as Baghdad, retrieves the FOUP from the conveyor and places it onto a transition table. Third, the Niryo One robot, referred to as Cairo, extracts individual wafers from the FOUP and transfers them to an alignment station. At the alignment station, a fixed overhead vision sensor captures the wafer image, and a trained convolutional neural network estimates the wafer notch orientation. The aligner then rotates the wafer to a corrected reference angle before Cairo transfers the wafer to a simulated processing chamber.

The project integrates several concepts covered in the Industrial Robotics (IE574) course, including robot work cell design, motion planning, end-effector control, signal-based task coordination, machine vision, and learning-based perception. Robot motion using Bezier path-planning was designed to avoid collisions while maintaining repeatable pick-and-place behavior. The system also uses asynchronous signal passing between robots and external Python-based inference to coordinate the overall process. This modular structure allows each subsystem to operate independently while remaining synchronized through event-based communication.

The main contribution of the project is the development of a multi-process automation pipeline that demonstrates the transition from bulk material transport to precision wafer handling. Rather than focusing on a single isolated robot task, the simulation emphasizes the interaction between multiple devices, the timing of handoffs, and the verification of wafer orientation prior to processing. The result is a realistic representation of challenges encountered in semiconductor automation, including payload transfer, wafer positioning, orientation uncertainty, and system-level coordination.

# Motivation

Front Opening Unified Pods, or FOUPs, are important equipment when it comes to cleanroom semiconductor manufacturing. These are front opening enclosures which are used to transport wafers safely and efficiently during the semiconductor manufacturing process. They are built for interaction with robots, allowing robots to autonomously handle the wafers as they go through the phases of their manufacturing processes, further protecting them from contamination and reducing the need for manual labor. Wafers must undergo hundreds of steps during the semiconductor manufacturing process. The integration of autonomous robotic systems across each stage of the process eliminates manual handling, reducing cycle time variability and the risk of human error, both of which are critical considerations in high-precision semiconductor manufacturing environments.

The integration of autonomous robots into the semiconductor manufacturing line can take a multitude of paths. In this case, the optimization of FOUP handling across a variety of surfaces was explored. In current applications, an individualized FOUP unloading robot would deliver the FOUPs to different handling robots in a fabrication sector. To eliminate potential pinch points and make the transport more efficient, this simulation explores the use of a conveyor belt delivery system, which can be delicate and precise for the fragile wafers, while also allowing for more FOUPs to be transported in less time than with an individualized FOUP unloading robot interface.

The proposed solution will use two robots to manipulate the wafers through the simulated fabrication sector. To optimize the points between FOUP drop-off and wafer handling, a multi-jointed manipulator robot with a suction pad end delivers the FOUP to a platform on which a smaller, precision robot can extract each individual wafer, and place it into a motorized autonomous aligner, which utilizes machine vision to ensure wafer discs have a uniform notch direction. This is an essential step prior to moving the wafer onto any processing steps (like lithography) since misaligning the wafer can cause harm to both the product and the processing machine, some of which can cost millions of dollars. From here, the wafers exit the simulated section of the manufacturing process.

The simulation for the proposed solution will capture a single frame of a much larger picture. In a real fabrication facility, the cell modeled here (one FOUP delivery robot, one wafer handling arm, one aligner, one process chamber) would be replicated across an entire semiconductor wafer handling and processing floor, potentially hundreds of times over, each running the same sequence in parallel and feeding into shared downstream process tools. The signals that coordinate handoffs between robots in this simulation are conceptually identical to the equipment communication standards that real SCADA systems use to orchestrate thousands of concurrent material movements.

# Problem & Background

The modern fabrication facility operates on a scale where the thickness of a single human hair is considered a massive obstruction. As the industry pushes toward increasingly dense circuit patterns, the margin for error in material handling has effectively vanished. The central challenge lies in the transition between bulk transport and individual processing. When materials are moved through a facility, they are subject to kinetic forces and environmental variables that, if not mitigated by intelligent machinery, would lead to catastrophic yield loss. Traditional infrastructure often relies on sprawling, rigid architectures that lack the agility required for rapid, localized processing. Consequently, there is a burgeoning need for modular cells that can handle the transition from transit to operation with absolute fidelity.

This demand is not merely a question of mechanical precision. Modern-day semiconductor processing systems operate under strict cleanroom classifications where airborne particle counts are controlled to single digits per cubic meter. Any mechanical system introduced into this environment must itself be a non-contributor to contamination, meaning that the materials, lubricants, motion profiles, and even the vibration signatures of robotic handlers are subject to engineering scrutiny. The introduction of Atmospheric Robots and Overhead Hoist Transport (OHT) systems in the 1990s marked the first generation of facility-wide automation, but these systems were designed around fixed process layouts and batch sizes that are increasingly misaligned with the flexible, on-demand production models demanded by modern design cycles.

At the wafer level, each substrate may pass through over 500 discrete process steps before reaching final inspection, with each step introducing its own handling requirements, orientation constraints, and contamination risks. The FOUP (Front Opening Unified Pod) emerged as the industry's answer to inter-tool transport, providing a sealed, nitrogen-purged environment that isolates wafers from ambient air during transit. However, the handoff between the FOUP and the process tool remains one of the most mechanically complex and failure-prone events in the entire production flow, demanding coordination between load port mechanisms, atmospheric robots, and alignment systems that must operate in concert with sub-millimeter repeatability across hundreds of thousands of cycles.

Robust integration of autonomous agents into these environments is crucial. A robot in this field must do more than move from point A to point B. It must understand the spatial orientation of the payload in real time. This requirement stems from the fact that mechanical vibrations during transport can cause subtle shifts in the positioning of the cargo. Without a system capable of identifying these shifts and correcting them before the next phase of manufacturing begins, the entire sequence is compromised. This necessitates an approach where specific hardware is dedicated to distinct sensory and motor tasks, ensuring that no single point of failure disrupts the continuity of the production line.

Not all fabrication facilities have the structural luxury of accommodating overhead hoist transport systems. A significant portion of the global semiconductor supply chain still operates out of older buildings constructed before OHT infrastructure became an industry standard. These facilities face a fundamental architectural constraint: their ceiling clearances, floor load ratings, and inter-bay layouts were designed around manual cassette handling and early generation automated guided vehicles, not the monorail systems and pod carriers that define modern 300mm

wafer fabrications and handling. Retrofitting such a facility with full OHT capability is not merely a capital expenditure problem but it often requires partial or full production shutdowns, structural reinforcement of the production floor and ceiling grid, and a complete re-qualification of the cleanroom environment, a process that can take years and cost hundreds of millions of dollars. For many existing producing specialty semiconductors, power devices, or MEMS components, the economics of a full OHT retrofit simply do not justify the investment relative to the production volumes involved. This creates a class of facility that is permanently excluded from the logistics architecture that the industry has standardized around yet still faces the same fundamental material handling demands in terms of contamination control, orientation fidelity, and throughput consistency. Lateral transport architectures using conveyor systems and collaborative robotic cells offer a pragmatic middle path that can be installed incrementally, qualified zone by zone, and scaled without requiring changes to the facility superstructure. Rather than rebuilding the factory around the logistics system, the logistics system is rebuilt around the factory.

In a localized environment, the replacement of traditional overhead logistics with a streamlined lateral flow creates a new set of mechanical demands. The synchronization required to intercept a payload from a moving surface and transition it into a static processing state requires high-frequency data exchange between the transport hardware and the robotic manipulators. By delegating the heavy lifting and the fine-motor adjustments to specialized units, the system explored in the simulation achieves a level of "functional compartmentalization." This allows for higher throughput, as each component of the cell can reset for the next cycle while the subsequent robot is still completing its task. It transforms a linear bottleneck into a parallel-capable workflow, which is essential for scaling production without increasing the physical footprint of the cleanroom.

At the core of what makes this kind of system actually work is the combination of two things that, on the surface, seem like they would conflict with each other: speed and precision. Path planning handles the speed side of the equation. Rather than having the robot react to where the wafer is after it gets there, the system plans the trajectory in advance, so the arm is already moving in the right direction before the previous step has fully finished. That predictive quality is what keeps cycle times tight without sacrificing control.

The machine vision side handles precision. Once the wafer handling robots begin processing individual wafer, cameras can be used to verify orientation, confirm positions, and feed that information back to the robots, so any correction can happen before the wafer ever touches the process tool. It is essentially a second opinion built into the workflow that may seem insignificant that would otherwise only show up as yield loss three steps later. The combination of the two (plan the path, then verify the result) is what closes the loop between high-speed transport and the sub-millimeter tolerances that lithography tools demand.

What the simulation work demonstrates is that this does not have to be an exotic or prohibitively expensive architecture. By prototyping these interactions in a controlled virtual environment first, it becomes possible to stress-test the logic, identify the failure modes, and tune the timing before a single piece of hardware is commissioned. The end result is a system that is genuinely faster and more consistent than what traditional handling approaches can offer and one that scales reasonably well as chip designs keep getting more complex, and the tolerance for handling errors keeps shrinking.



# Method

## Process Environment and Conveyor Systems

The process begins at the material handling interface, where an automated ground-level conveyor system delivers Front Opening Unified Pods (FOUPs) into the robotic work cell from preceding facility sections. Positioned slightly above floor level, the conveyor provides an effective representation of automated semiconductor container delivery from an upstream process or storage area, while ensuring the FOUP arrives at a height reachable by the handling robot without requiring excessive vertical motion. This simplified the pick-up routine and reduced the likelihood of unreachable IK configurations. To manage flow and prevent bottlenecks, the conveyor is segmented into three distinct arrival zones, allowing up to three FOUPs to queue simultaneously. This buffering capability ensures that the primary handling robot always has immediate access to a container, maintaining a constant operational cadence without idling. Although geometrically simpler than a real semiconductor transport system, the conveyor effectively demonstrates a key industrial automation concept: the coordination of a robot with an external machine in a multi-process system, requiring the robot to wait for the FOUP to arrive at a designated pickup location before initiating its manipulation sequence.

Once a FOUP is staged in the final queue position, the heavy-lifting phase is handled by Baghdad, a Jaco manipulator equipped with a vacuum suction cup end-effector. Baghdad executes a precise pick-and-place operation, transferring the loaded FOUP from the conveyor to a central transition table that serves as the critical junction between the facility-wide transport system and the localized wafer handling cell. By utilizing a high-payload arm for this stage, the simulation accounts for the significant weight of fully loaded FOUPs while maintaining the millimeter-level accuracy required for the subsequent hand-off.

Individual wafer manipulation is then managed by Cairo, a Niryo One robot also equipped with a vacuum suction cup end-effector. Cairo extracts wafers one at a time from the staged FOUP and transfers each to a camera-based alignment station that calibrates orientation to sub-degree tolerances, which is a critical requirement in semiconductor processing. Following successful alignment, Cairo places each wafer into a dedicated processing box, which functions as a symbolic gateway representing the wafer's departure to specialized chemical or lithographic manufacturing steps. Once the FOUP is fully unloaded, it is transferred to a second conveyor positioned at ground level and in line with the transition table. The empty FOUP is animated sliding along this exit conveyor and off screen, where it is assumed to be reloaded with fresh silicon wafers and re-entered into the upstream supply chain, allowing Baghdad to begin loading the next FOUP and completing the cycle.

## Wafer Handling and Alignment: Cairo and Moscow

Cairo operates on a coroutine-based control architecture within CoppeliaSim's non-threaded script framework. Its main sequence runs as a Lua coroutine resumed each simulation step through `sysCall_actuation`, allowing the robot to yield mid-sequence while the IK solver continues running in the background. Motion is handled in Cartesian space via `sim.moveToPose` with a damped least-squares IK solver, and a known-good gripper-down orientation quaternion is captured at the start of each cycle to ensure consistent end-effector orientation regardless of IK

solution drift. The entire sequence is driven by signal-based handshaking with no direct script-to-script calls, making the system straightforward to extend with additional stations or robots. Cairo waits indefinitely for a FOUP\_Ready\_for\_Unload signal broadcast by Baghdad before beginning. Upon receiving it, Cairo performs two wafer cycles per FOUP, descending into the box to pick a wafer via simulated vacuum suction, transporting it to an aligner (Moscow), and placing it with a randomized Z-axis rotation to simulate unknown arrival orientation.

Wafers are represented by a single Notched\_Wafer shape featuring an industry-standard notch for orientation detection (Kim, J., 2018). Rather than cloning the object each cycle, the same shape is reused by toggling its visibility layer and re-parenting it between the suction cup attachment point and the aligner chuck as needed. Alignment is handled by an external Python inference server connected via the ZeroMQ Remote API. A fixed overhead vision sensor captures the wafer on the aligner, and a trained 1D convolutional neural network samples a 360-point edge brightness profile at 95% of the wafer radius, outputting the notch angle as a (sin, cos) pair to avoid angular discontinuity. Trained on 360 synthetic CoppeliaSim images, the model achieves a mean angular error below 1° after 50 epochs. Cairo triggers this pipeline by setting an align\_trigger signal and resumes only after receiving align\_done, confirming the aligner motor has reached the target angle within 1° tolerance. Once the alignment is done, Cairo moves back from the home position, where it rests while waiting on the align\_trigger signal, to the same drop off position where it placed the wafer on the motor aligner. Cairo then picks up the now aligned wafer, and via IK, travels to the chamber where it places the wafer in the same position every time. Once both wafers are processed and deposited into the chamber, Cairo broadcasts FOUP\_Unloaded to release Baghdad for its next cycle, then animates the empty FOUP sliding along the exit conveyor before removing it from the scene.

## **FOUP Unloading: Baghdad**

The robot selected for the pick-and-place task was the Kinova Jaco robotic arm, a lightweight assistive manipulator designed for precise object manipulation in compact environments. Although the Jaco is commonly associated with assistive applications, this project used it as a standalone robotic manipulator operating without human intervention. Its six-degree-of-freedom serial-link structure allowed the end effector to move through three-dimensional space while also controlling its orientation, which was essential for navigating between the FOUP, conveyor, wafer pickup area, alignment station, and placement location within the simulated semiconductor work cell.

The Jaco arm's compact design and moderate performance capabilities made it well suited for the scale of the simulation. With a maximum reach of approximately 900 mm, a total weight of about 5.2 kg, and a continuous payload rating of approximately 1.3 kg at full range or 1.6 kg at mid-range, the robot was appropriate for tasks requiring controlled, repeatable motion rather than heavy payload handling (Kinova, n.d.). While it would not be ideal for full-scale industrial FOUP transport, it effectively represented a lab-scale semiconductor automation process focused on wafer transfer, precise positioning, and coordinated pick-and-place operations.

To control the robot's motion in CoppeliaSim, inverse kinematics was used to guide the Jaco's end effector through each task location. The robot model used an IKtip to represent the



actual end-effector frame and an IKtarget to define the desired position and orientation. Rather than manually calculating each joint angle, target poses were assigned to the IKtarget, and CoppeliaSim's inverse kinematics solver calculated the joint configurations required for the IKtip to follow those targets (Coppelia Robotics, n.d.). This approach created a smooth and adaptable control method, allowing the robot to move reliably through the semiconductor workflow while maintaining realistic task-space motion.

There are two ways for path planning for robotics (Bulut, 2022):

1. Bezier Curve
2. Ferguson Curve

The formula for Bezier Curve is as follows:

$$r(t) = (1 - t)^3 P_0 + 3(1 - t)^2 t P_1 + 3(1 - t) t^2 P_2 + t^3 P_3$$

To calculate the way points using Bezier Curve; four points are needed.

P0: Starting Point or Home Coordinate

P3: Ending Point or Desired Coordinate

P1 and P2 are considered to be control points. They curve the robot path up and down depending on the location.

$$r(t) = [P_0 \ P_2 \ P_3 \ P_1] = \begin{bmatrix} P_{0x} & P_{2x} & P_{3x} & P_{1x} \\ P_{0y} & P_{2y} & P_{3y} & P_{1y} \\ P_{0z} & P_{2z} & P_{3z} & P_{1z} \end{bmatrix} \begin{bmatrix} -1 & 3 & -3 & 1 \\ 3 & -6 & 3 & 0 \\ -3 & 3 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} t^3 \\ t^2 \\ t \\ 1 \end{bmatrix}$$

Once the points are chosen, matrix multiplication is performed as indicated above to obtain the coordinates of the robot from time (0 – 1) with 0.1 step size.

To calculate the way points using Ferguson Curve; four points are needed.

P0: Starting Point or Home Coordinate

P3: Ending Point or Desired Coordinate

P1 and P2 points are chosen so that the tangent corresponds to the path that the robot should travel to the desired coordinate.

$$T0 = P2 - P0$$

$$T1 = P1 - P3$$

$$r(t) = [P_0 \ P_2 \ T_0 \ T_1] = \begin{bmatrix} P_{0x} & P_{2x} & P_{3x} & P_{1x} \\ P_{0y} & P_{2y} & P_{3y} & P_{1y} \\ P_{0z} & P_{2z} & P_{3z} & P_{1z} \end{bmatrix} \begin{bmatrix} -1 & 3 & -3 & 1 \\ 3 & -6 & 3 & 0 \\ -3 & 3 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} t^3 \\ t^2 \\ t \\ 1 \end{bmatrix}$$

Once the four points are chosen, T0 and T1 are calculated. Then the matrix multiplication is performed to obtain the coordinates from time (0 – 1) with 0.1 step size.

# Main Findings

## **CoppeliaSim Physics**

One of the most significant findings from the simulation was that the interaction between the robot, conveyor, and FOUP depended heavily on the object dynamics and respond-ability settings in CoppeliaSim. Although the FOUP appeared to be a simple object to transport, its behavior in the scene created several challenges during the pickup and placement process. The main issue was that the FOUP needed to respond physically to the conveyor while also being stable enough for the Jaco robot to grasp the suction pad. When the dynamic setting was turned off too early, the FOUP did not behave like a realistic transported object and instead spawned above the conveyor and hovered in place. This made the conveyor stage unrealistic and prevented the system from accurately demonstrating material delivery.

These challenges highlight a broader limitation of physics-based simulation that becomes critical when transitioning to real robotic systems. In CoppeliaSim, object dynamics are controlled through discrete parameters (static flags, respondability toggles, and mass properties) that are switched programmatically at precise moments in the sequence. In a real manufacturing environment, no such clean state transitions exist. A physical FOUP sitting on a conveyor is subject to continuous inertial, frictional, and vibrational forces simultaneously, and a robot gripper or suction cup must account for all of them during the grasp. Parameters such as conveyor belt surface friction, FOUP base compliance, suction cup seal integrity under lateral load, and gripper approach velocity all interact in ways that simulation either approximates coarsely or ignores entirely. If a real Jaco arm attempted to grasp a FOUP using the same timing and speed profiles developed in simulation, small discrepancies in contact force or surface friction could cause the FOUP to shift, tilt, or fail to seal under the suction pad, leading to a dropped payload.

## **Robot Design and Limitations**

The pick-and-place task also required several design iterations. The team tested multiple robot models before selecting the Jaco arm because many of the available robots were unable to reach, attach to, or lift the FOUP successfully. Even after selecting the Jaco, the FOUP's dynamic and respond-able properties had to be carefully managed. Leaving dynamics fully active allowed the FOUP to interact properly with the conveyor, but it made the suction-based pickup unreliable. Turning dynamics off permanently made pickup easier but caused unrealistic hovering and removed the physical interaction needed for conveyor transport. The final solution was to modify the script so that the FOUP remained dynamic and respond-able while traveling on the conveyor, then switched dynamics off only after the vacuum on the Jaco suction pad was activated. This allowed the FOUP to behave realistically during delivery while still enabling a successful suction-based pickup.

Another issue involved robot joint movement and reachability. During early testing, the Jaco arm could not consistently reach all required task locations while maintaining a useful end-effector orientation. Some target poses placed the robot near the limits of its workspace, and

certain movements caused awkward joint configurations that interfered with smooth inverse kinematics motion. These limitations were addressed by rearranging the work cell layout and reducing the overall footprint of the system. By bringing the conveyor, FOUP pickup location, and placement area closer together, the robot could complete the task with fewer extreme joint movements and more stable IK behavior.

## **Vision Limitations**

Another major finding was during the development of the wafer alignment system, where the vision sensor initially failed to produce a usable signal for notch detection. The first iteration of the simulation used a low default camera resolution of  $16 \times 16$  pixels, which was insufficient to resolve the small geometric cutout on the wafer edge. At this resolution, the notch occupied only a fraction of a pixel in the captured image, making it indistinguishable from general edge noise caused by the overall geometry of the wafer shape in the simulation. When the edge brightness profile was plotted, multiple shallow dips of similar magnitude appeared distributed around the circumference, and no single feature could be reliably identified as the notch. The solution was twofold. First, the vision sensor resolution was increased to  $256 \times 256$  pixels, providing sufficient spatial detail for the notch to appear as a distinct dark region against the bright wafer surface. Second, the edge sampling radius in the profile extraction algorithm was tuned to 95% of the detected wafer radius, ensuring that the sampling ring passed through the notch geometry rather than slightly inside or outside it. With these adjustments, the resulting edge profile showed a single sharp dip at the notch location while remaining flat everywhere else, which the 1-D CNN was then able to detect reliably.

This issue has direct parallels in real semiconductor manufacturing environments. Industrial wafer aligners rely on optical sensors, typically through-beam or reflected light sensors rather than cameras. The same fundamental principle applies: the sensing resolution must be matched to the feature size being detected. A 300mm silicon wafer has a notch approximately 1mm deep and 2.5mm wide at the edge. If a camera-based system were deployed with insufficient resolution or poor focal depth, the notch would be lost in optical noise, leading to misaligned wafers entering the process chamber. In thin film deposition or lithography steps, even a few degrees of rotational misalignment can cause pattern overlay errors that propagate across the entire wafer, resulting in yield loss across hundreds of dies. This underscores the importance of sensor selection, calibration, and resolution validation as a critical step in any vision-guided alignment system before deployment in a production environment.

These findings demonstrated that successful robotic simulation requires more than placing objects and assigning motion commands. The physical properties of the FOUP, the timing of the suction activation, the conveyor height, and the robot's reachable workspace all had to be coordinated for the system to function correctly. The final implementation showed that maintaining FOUP dynamics during conveyor motion and disabling them only during suction attachment provided a practical balance between realistic material behavior and reliable robotic manipulation.

For future iterations of this simulation, it would be valuable to incorporate more realistic physics modelling, including friction coefficients matched to real conveyor materials, suction force curves based on vacuum pump specifications, and dynamic gripper compliance modelling.

CoppeliaSim supports multiple physics engines which offer significantly higher fidelity contact modelling than the default Bullet engine used here. Running the same sequence under these engines and comparing FOUP stability outcomes would provide a more honest estimate of how robust the pick-and-place routine would be under real-world conditions and would surface failure modes before any physical hardware is commissioned.

# Evaluation

The system was evaluated based on its ability to demonstrate a complete integrated robotic workflow involving multiple robots, multiple processes, and multiple tasks. The primary evaluation criteria were successful FOUP transport, successful FOUP transfer by Baghdad, successful wafer extraction by Cairo, successful wafer alignment using machine vision and machine learning, and successful transfer of the wafer to the simulated processing chamber.

From a system integration perspective, the simulation successfully demonstrates a multi-stage semiconductor handling process. The conveyor delivers the FOUP to the pickup area, Baghdad transfers it to the transition table, Cairo extracts wafers, the aligner corrects wafer orientation, and the wafer is delivered to the processing box. This satisfies the project requirement for a unified automation framework involving multiple tasks and processes rather than a single isolated robot operation.

The motion planning evaluation focused on whether each robot could complete its assigned movements without collisions or unreachable poses. Baghdad's task required larger-scale movement between the conveyor and the table, while Cairo's task required smaller, more precise movements between the FOUP, aligner, and processing box. The use of planned intermediate positions helped avoid collisions and improved the repeatability of the robot motion. The need for these waypoints demonstrated the practical importance of path planning in cluttered robotic environments.

The vision and machine learning subsystem was evaluated by the accuracy of wafer notch detection and the ability of the aligner to rotate the wafer to the desired orientation. The CNN-based method achieved a mean angular error below one degree in training, which is suitable for the simulated alignment objective. The alignment process also demonstrated successful communication between the CoppeliaSim environment and an external Python inference server through the ZeroMQ Remote API.

The overall system showed strong modularity. Each major subsystem (conveyor, Baghdad, Cairo, aligner, and vision server) could be developed and tested separately, then integrated through signal-based communication. This architecture improved reliability and made the work cell easier to extend. For example, additional FOUPs, additional processing chambers, or an inspection station could be added without requiring a complete redesign of the control logic.

However, the simulation also has limitations. The FOUP and wafer handling behavior is simplified compared with a real semiconductor cleanroom. Real FOUPs contain many wafers, require highly precise mechanical interfaces, and must operate under strict contamination control requirements. The suction interaction and wafer transfer are also idealized relative to real vacuum grippers and wafer end-effectors. In addition, the machine learning model was trained on synthetic images from the simulation environment, so its accuracy may not directly transfer to real camera images without additional domain adaptation.

Despite these limitations, the project successfully validates the feasibility of the proposed robotic workflow at the simulation level. It demonstrates that conveyor-based FOUP delivery, robotic FOUP handling, robotic wafer extraction, and machine learning-based wafer alignment can be combined into a coordinated automation cell.



# Conclusion

This project successfully demonstrated the design and simulation of an integrated robotic semiconductor handling cell in CoppeliaSim. The system combined a floor-level conveyor, a Kinova Jaco robot named Baghdad, a Niryo One robot named Cairo, a wafer alignment station, machine vision, and a CNN-based notch detection model into one coordinated automation workflow. Rather than focusing on a single robot task, the project modeled a multi-stage process in which a FOUP is delivered by conveyor, transferred by the Jaco robot, unloaded by the Niryo One robot, aligned using vision-based feedback, and moved into a simulated processing chamber.

A major outcome of the project was the successful application of several core industrial robotics concepts. The Jaco robot's six-degree-of-freedom structure and CoppeliaSim's IKtip and IKtarget framework allowed the team to implement inverse kinematics for controlled task-space motion. Bezier-based path planning and intermediate waypoints helped reduce collision risk and improve repeatability. The use of signal-based communication allowed the conveyor, robots, aligner, and external Python inference server to operate as independent subsystems while remaining synchronized throughout the workflow.

The project also revealed several practical challenges involved in robotic simulation. The most significant issues involved the FOUP's dynamic and respond-able properties, suction-based pickup, robot reachability, and work cell layout. These challenges required several design iterations, including testing multiple robots, modifying object dynamics during suction activation, rearranging the work cell, and reducing the system footprint. Solving these issues strengthened the final simulation and showed that successful automation depends not only on robot motion, but also on object behavior, machine interaction, timing, and system-level coordination.

Overall, the completed simulation validates the feasibility of a lab-scale semiconductor automation cell that integrates material delivery, robotic FOUP handling, wafer manipulation, and learning-based wafer orientation correction. While the model simplifies some real cleanroom conditions, such as FOUP payloads, contamination control, and production-grade wafer handling, it provides a strong representation of how robotic systems can be coordinated for semiconductor manufacturing applications. Future improvements could include more realistic FOUP geometry, sensor-based conveyor feedback, expanded collision detection, additional processing stations, and further testing of the machine vision model under more varied simulated conditions.





# Contributor Acknowledgments

| Team Member      | Contributions                                                    |
|------------------|------------------------------------------------------------------|
| Daniel Ko        | Baghdad code and FOUP Conveyor, Report                           |
| Mahad Mohamed    | Baghdad code and FOUP Conveyor, Report                           |
| Sanjidah Hossain | Cairo and Aligner code, Report                                   |
| Ruben Bonilla    | Cairo and Aligner code, Report                                   |
| Abrea La Grone   | Simulation Environment Setup, Report,<br>PowerPoint Presentation |
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